B204 App & Web Development — Individual Project Report

NexShop: An E-Commerce Web Application

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Module: B204 App & Web Development **Submission Date:** March 26, 2026

GitHub Repository: https://github.com/Saaqibx77/b204ecommercewebapp

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1. Introduction

This report consists of all information about design front end, backend and deployment information of the e-commerce application as a part of B204 App & web Development Course.

This application about basic functionality as per the project requirements like search for products, viewing product details, add items to the cart, complete purchase

using dummy paypal account.

The application is built using front end tools HTML,CSS,Javascript also used Node.js, Express.Js used for back end, MongoDB is used for data storage.

The project is used to cover full web site development and payment integration with good UI features.

2. Project Overview

Project follows three tier client server architecture.

Used Front End using HTML, CSS, JavaScript

Technology used:

- **HTML,CSS,JavaScript : Used for Front end Development it is used because it is used to design the structure of the web page and designing in CSS with help of that we have tried to make it more attractive, user friendly and easy to use.**

  It also helps to make the website lightweight that requires less amount of memory for storage and it is easy to understand and implement as compared to other complex UI Development tool. As part of implementation we have designed the front end system user friendly.
- **Node.js , Express.js:** Technology which is an important part of the project as it provides easy connectivity with the database mongodb which is used to store the database. Advantage of using Node.js is that it is used to provide JavaScript based development system which can be readily connected with our front end. Which is an essential part of our development. Express.js is also used with Node.js to provide RESTful APIs in support of JSON.
- **MongoDB : it is used as our data storage system MongoDB is good for e commerce type data storage it provides flexible way to store our data.**

  MongoDB provide us easy to used document oriented system which is easy to use.

  It has flexibility of changing attribute count based on the requirement of the data storage. It is used to avoid some strict relational constraints. Attributes can be easily mapped to JSON like structure for the file.

3. Front-End Design

The Front end design is done with the help of easy to use integration of CSS with the basic html file to provide attractive look to the website and also the web contents will be more easy to use for the end user point of view.

Design System

A comprehensive CSS design system was created in style.css using CSS custom properties and variables with values and designed according to modern requirements of the website development.

```
.root

{

}
```

html

{

}

h1{

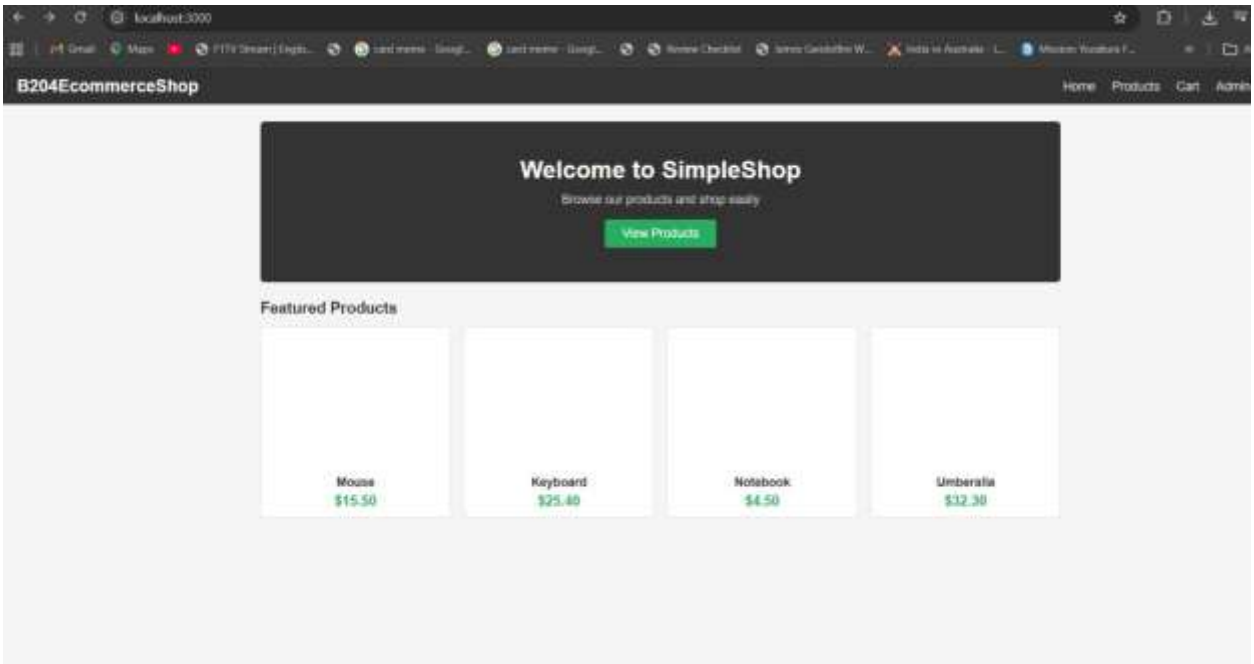
}

The design provides decent color combination with excellent and smooth effects that provides modern way of representing websites having CSS with excellent artistic effects.

All Page Information

(home.html) Page

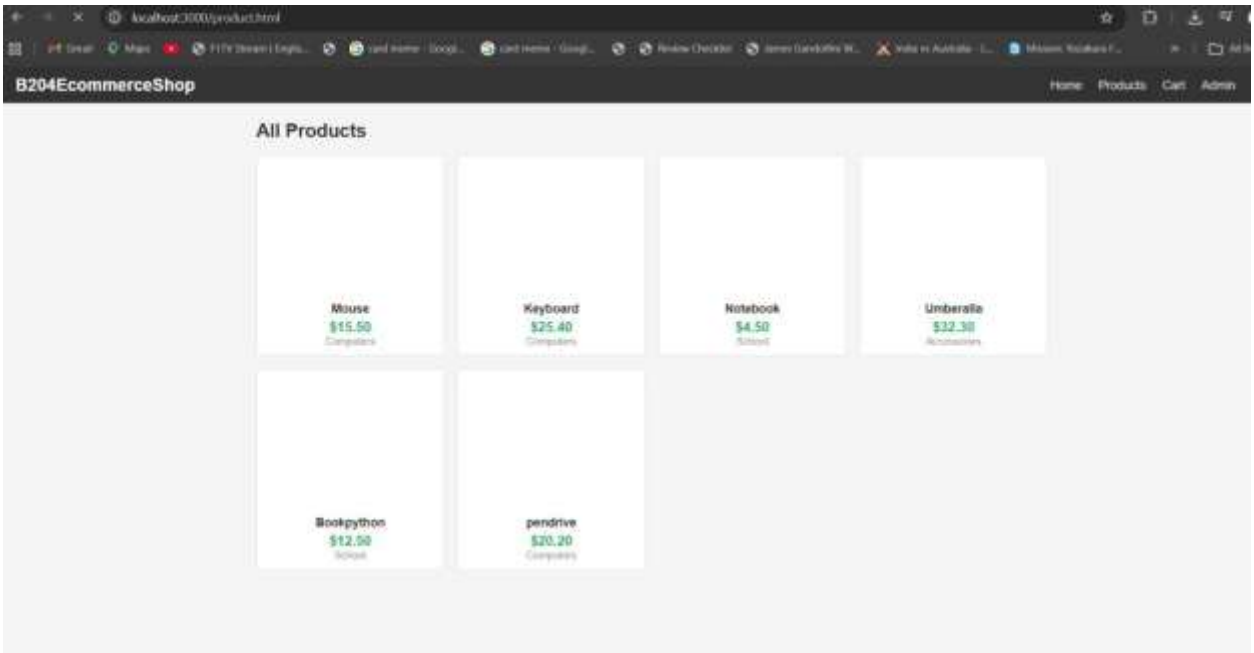
it is the first page to start with which is Landing page provides select product option and the buttons easy to navigate with cards provided for Secure payments also system required to display the products with are latest or recently used.



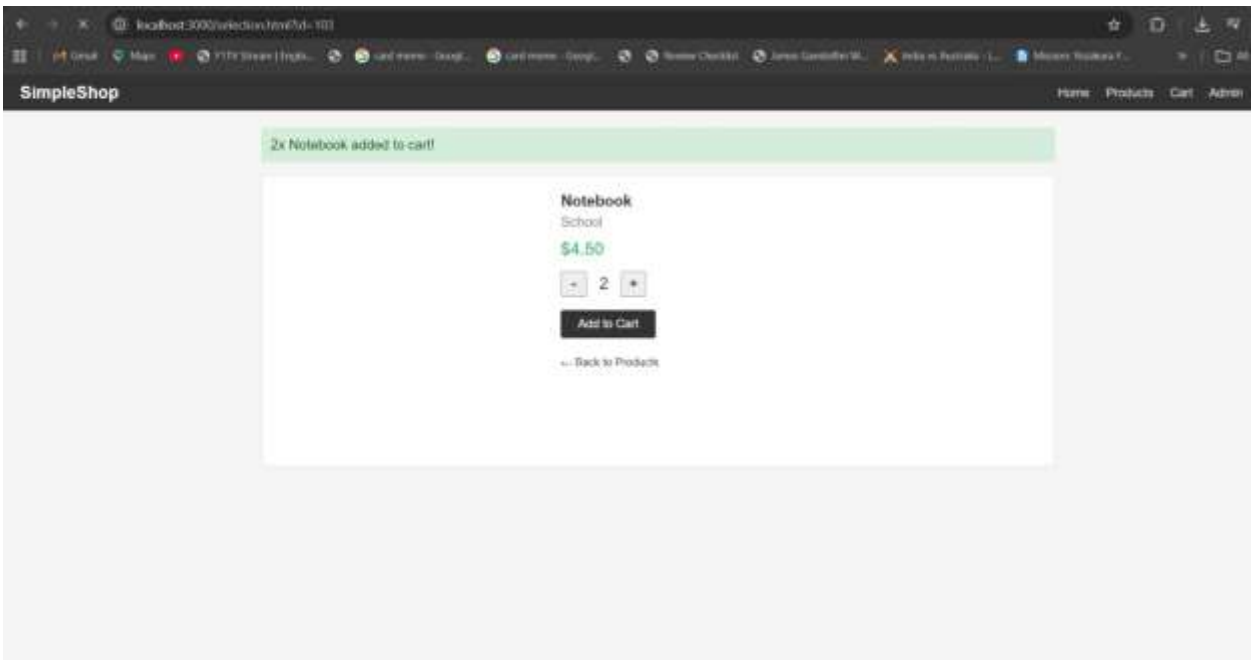
Product information page (product.html)

This page display all the products in the Grid Layout created using CSS Grid Layout

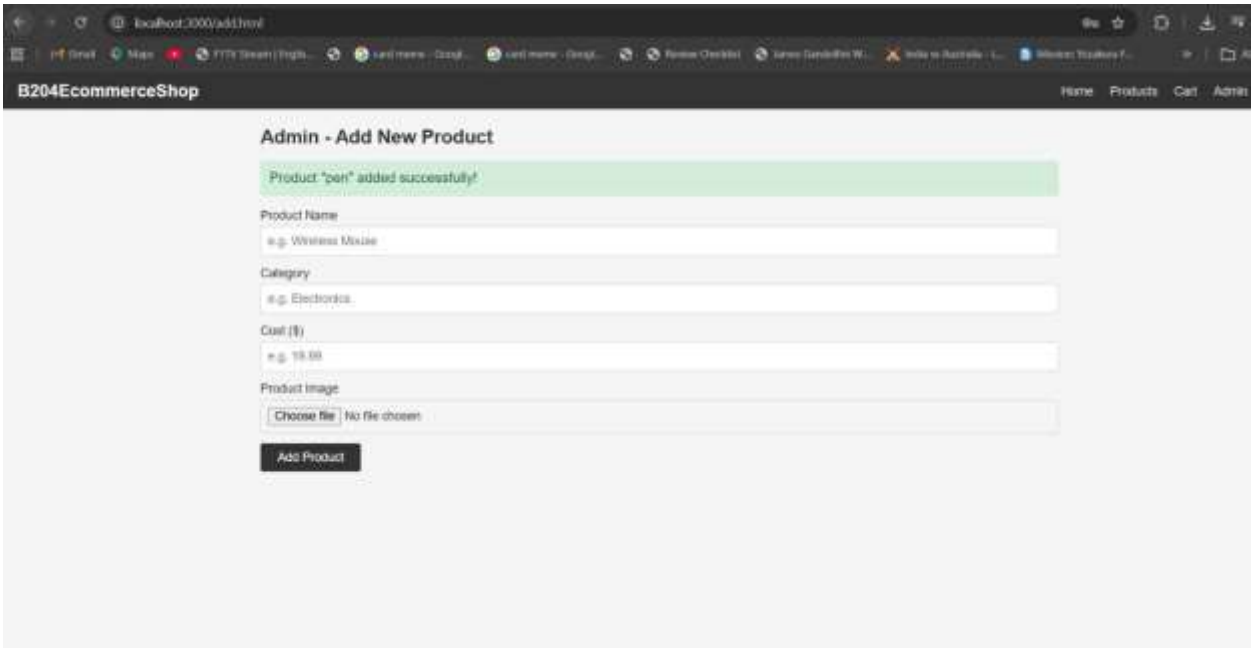
Each Card is created using a product image, category, name of the product, cost and an option to add to cart. A search option is provided on top of the page which is used to filer the products search by name.



Product Selection page(selection.html) : it shows a two column layout with product image is visible on the left side and an option to select the quantity on the right which can be increased or decreased. Also the price will keep on changing automatically when clicking on plus(+) or minus(-) button in the interface.

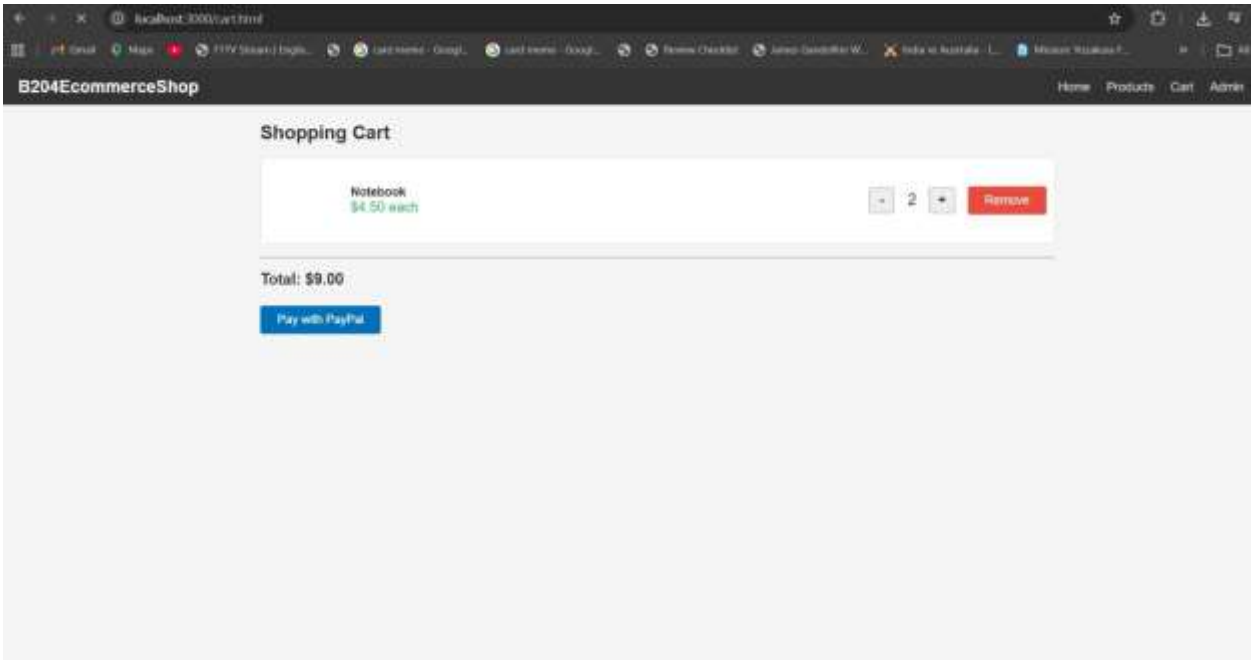


Add Product Page (add.html): Contains system to provide text input for entering the name , text input for brief description an option to add an image with the given URL a number field to input the price and a simple drop down list to select the category from the available categories of the product. Some Client side validations are done using simple HTML 5 required attribute.

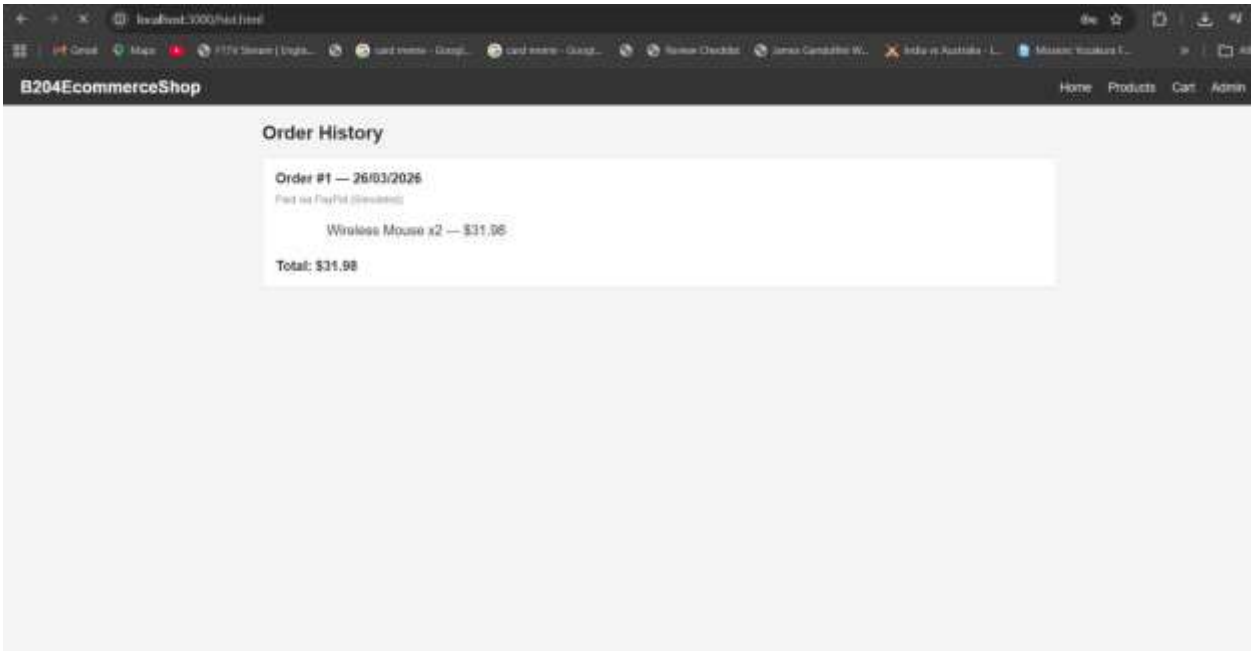


Checkout Page (cart.html): A web page that uses two column Grid layout. In the left side of the column product name are listed with the quantity selector option and option to remove which is given by the cancel button.

In the right side order summary is visible and a button which is showing to pay using paypal. Cart data is simply maintained locally using shared app.js utility.



History Page (hist.html): This page display all completed orders in a tabular form with the list of items showing orderid, paypalid of payment, date, items name ordered.



Responsive Design

Media query is used to adapt the layout according to device requirements the website adapts according to mobile or large screen systems. Layout and design is set according to the requirements of the web page so that all cards and all data looks good.

JavaScript Interactivity

System is provided with app.js file which is used to provide reusable requirements.

- **Handling Cart :** it is provided using addSomeCart(), cleanFromCart(), findAmtTotal() all backend with support of local storage.
- Some Api Functions are provided as apiGet(), apiPost() which are used for various error handling functions.

4. Back-End Development

4.1 Setting up Server

According to the server requirements server.js provides the entry point and express used as the middleware which worked with Node.js.

```
app.use(cors());                //is is used for resource sharing
app.use(express.json());        //it is used to Parse for JSON
app.use(express.urlencoded({extended:true}));//It is used to parse the data
app.use(express.static('public'));  //it is used to help in frontend files
```

Modularization is used to route into required files or directories. Environment variable with the help of dotenv is used to keep sensitive information or configuration like paypal user id and the password which can be used outside the required codebase.

Seed function is used to automatically get the data of six sample products when the first time website is run this will help in making the website directly usable without any user activity.

4.2 Database Schema Designed using MongoDB

Two Schemas are designed in the data layer of the system.

First Schema (Product)

```
const prodSchema = new mongoose.Schema({
  pnm:       {type:String,required:true},
  descr: {type:String,required:true},
  cost:  {type:Number,required:true,min:1},
  prdimg:      {type:String,default:'defaultimage.png' },
  prodcat:    {type:String,required:true},
  dateofcreation:   {type:Date,default:Date.now}
});
```

Second Schema(Order):

```
const myorderSchema=new mongoose.Schema({
  myitems:        \[{ productId:ObjectId,iname:String,cost:Number,pqty: Number}],
  totalBillValue:{type:Number,required:true},
  paypalId: {type:String,default:''},
 buyerEmail: {type:String, default:'purchase@myshop.com'},
  dateOfOrder:     {type:Date,default:Date.now}
});
```

4.3 Search Implementation

The product search using MongoDB which provides regex matching which is used on the name field. It is used to provide the case insensitive information for that it uses $regex option in the get() with anync a flag is used with the value of 'i' which indicate that the searching can be done with the help of easy system search and helps to provide real time search with case insensitive information helpful for all type of users for easy search option which is good feature of searching in the website.

5. Payment Integration

PayPal payment System in the website

Application is used to provide simple paypal payment system which is not real but provides a realistic view of the payment for the client. The implementation is easy and provides the required payment option which is easy to use for end user point of view.

A secure system is simulated using Client Id in the paypal uses POST and .env to obtain the required token. When the option of payment comes user can select the make payment using paypal at the front end and POST is called and an order is created with the required total amount. After the user confirms the purchase the front end calls the POST and order transaction is finalized and virtually payment is made. This is a smooth and easy way of paypal payment simulation.

Checking the status for Success or the Failure

- In case of success a screen displays a order confirmation message with the required paypal order Id and an option by which we can see the ordered items and the option to continue shopping.
- In case of failure a message is displayed informing the user that some failure occurred also a checkout page will remain there so that user or the buyer has an option to retry the payment.

6. Database Schema

In the database using MongoDB B204Ecommerce two collections are created.

First Collection: ProductInfo This collection contains the information about the information about product data. Documents are created which is used to store product name , cost of the product, description or information about the product. URL of the image, product category information an index is also created on the product name so the searches will be more efficient.


```
[
  {
    "id": 101,
    "name": "Mouse",
```

"category": "Computers",
"cost": 15.50,
"image": "/uploads/mouse.jpg"
},
{
"id":102,
"name":"Keyboard",
"category": "Computers",
"cost": 25.40,
"image": "/uploads/keyboard.jpg"
},
{
"id":103,
"name":"Notebook",
"category": "School",
"cost": 4.50,
"image": "/uploads/notebook.jpg"
},
{
"id":104,
"name":"Umberalla",
"category": "Accessories",
"cost": 32.30,
"image": "/uploads/umberalla.jpg"
},
{
"id":105,
"name":"Bookpython",

```
    "category": "School",
    "cost": 12.50,
    "image": "/uploads/bookpython.jpg"
  },
  {
    "id":106,
    "name":"pendrive",
    "category": "Computers",
    "cost": 20.20,
    "image": "/uploads/pendrive.jpg"
  }

]
```


Second Collection:


```
[
  {
    "id": 101,
    "items": [
      {
        "productId":1,
        "name": "Mouse",
        "cost": 15.50,
        "image": "/uploads/mouse.jpg",
        "quantity": 3
      }
    ],
    "total":"46.50",
```

"email":"newcust@myweb.com",

"address":"151, gisma univ",

"paymentMethod":"PayPal",

"date": "2026-03-25T21:07:25.971Z"

}

]

OrderInfo this collection is used to store information about all the successful purchase information. In the case of each order information is stored as an array of purchased items in the array each item information is stored as name of product, cost of the product, quantity of each product purchased, total amount of the items purchased also the image url information path of the image .

MongoDB, JSON Structures are used over Structured Query Language as MongoDB provides document like structure and also provides schema based freedom to work with as new fields can be easily added in the same collection without any worry about database migration.

7. Docker & Deployment Setup

Running in the Local System for this the requirements are Node.Js v15+

Project is developed in Windows 11 with Node , Browser to test the web application

```
cd b204projececom
npm install
node server.js
```

to run Open using http://localhost:3000

Dockerfile

Dockerfile will be provided to run the project on different system with the different requirements project is developed in Windows 11 and the Docker information can be run on different systems. Mac Systems may cause compatability issues tried best to run on mac as well as Linux

Environment Variables

PORT=3000
MONGODB_URI=mongodb://localhost:27001/B204Ecommerce

The application is made flexible in case of non availability of MongoDB automatically switches to local memory in the system so that it can run easily even without MongoDB allowing the application to be run easily without database installation.

8. GitHub Repository

- **Repository URL:** https://github.com/Saaqibx77/b204ecommercewebapp

- **Visibility:** Public

File Structure:

```
b204ecommerce/
    data/
        -cart.json
        -orders.json
        -products.json
    public/
        uploads/
            -mouse.jpeg
            -keyboard.jpeg
            -phonebook.jpeg
            -Umberallay.jpeg
        -add.html
        -cart.html
        -hist.html
        -home.html
        -product.html
        -selection.html
        -style.css
    -package.json
    -README.md
    -server.js
```

9. Conclusion

Our Ecommerce website provide web application that helps in product listing an index page which is easy to use with that all the pages are smoothly linked with the appropriate hyperlinks provides easy navigation to different pages, all the necessary product information is provided with the listing of product information like cost, product category and other required information of the website paypal simulation integration is provided so that payment gateway in virtual environment is implemented, cart is created with proper addition of the products and option of adding and deleting the products from the cart. The project is fully functional with complete project requirements are provided so that it can run on local system with the help of Node.js and with the basic database requirements and also with the help of Docker

10. References

https://expressjs.com/ *express.js express js related information how to install and all*

https://expressjs.com/en/resources/community.html community information of express.js

https://nodejs.org/docs/latest/api/ all information about the node.js and for the implementation learning and codebase help

https://nodejs.org/docs/latest/api/assert.html used for node:assert module information used in the project development

https://nodejs.org/en/download for downloading the latest nodejs version having Long term support download windows installer(.msi) file

https://www.mongodb.com/docs/ all the document information for understanding the mongodb and document structure creation.

https://mongoosejs.com/docs/

mongoose documentation to learn all information about mongoose that is helpful in backend implementation